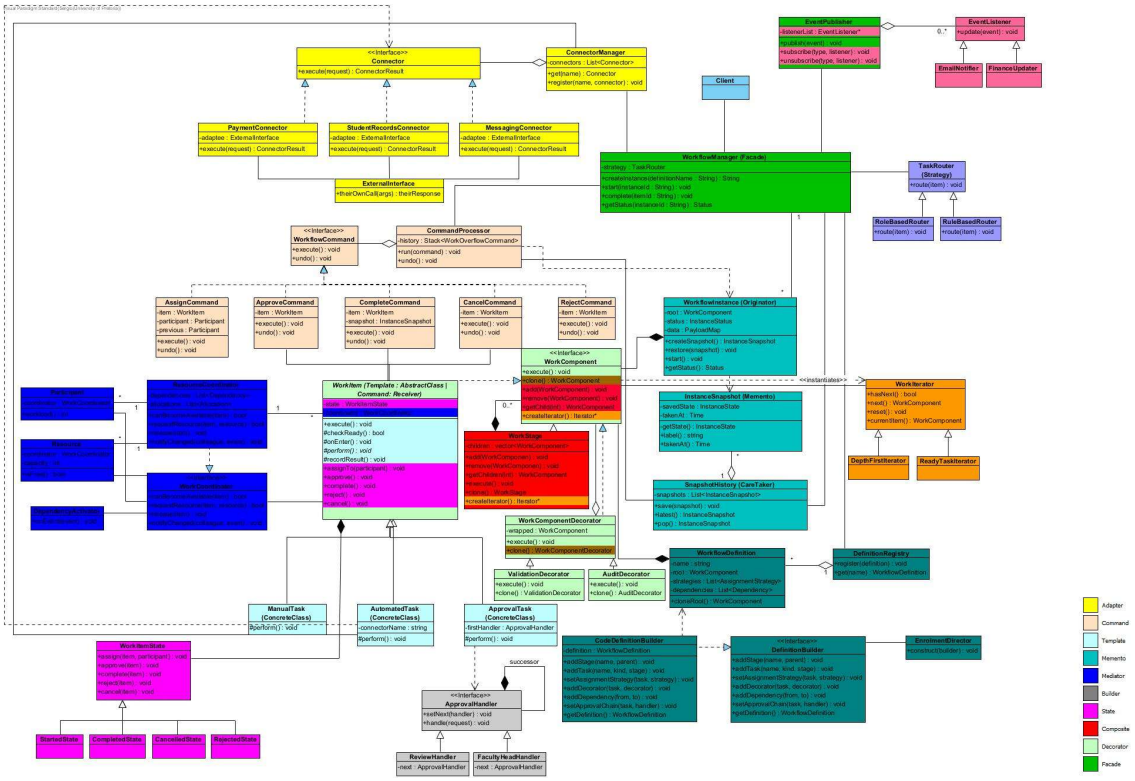
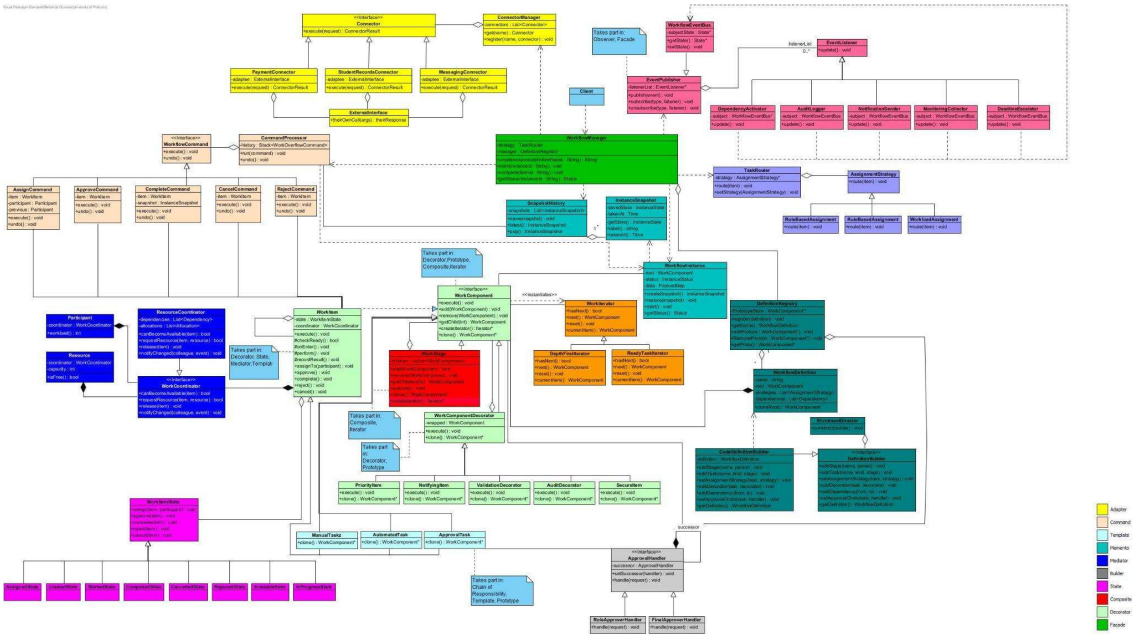
Task 7

Initial Class Diagram & Changes made:

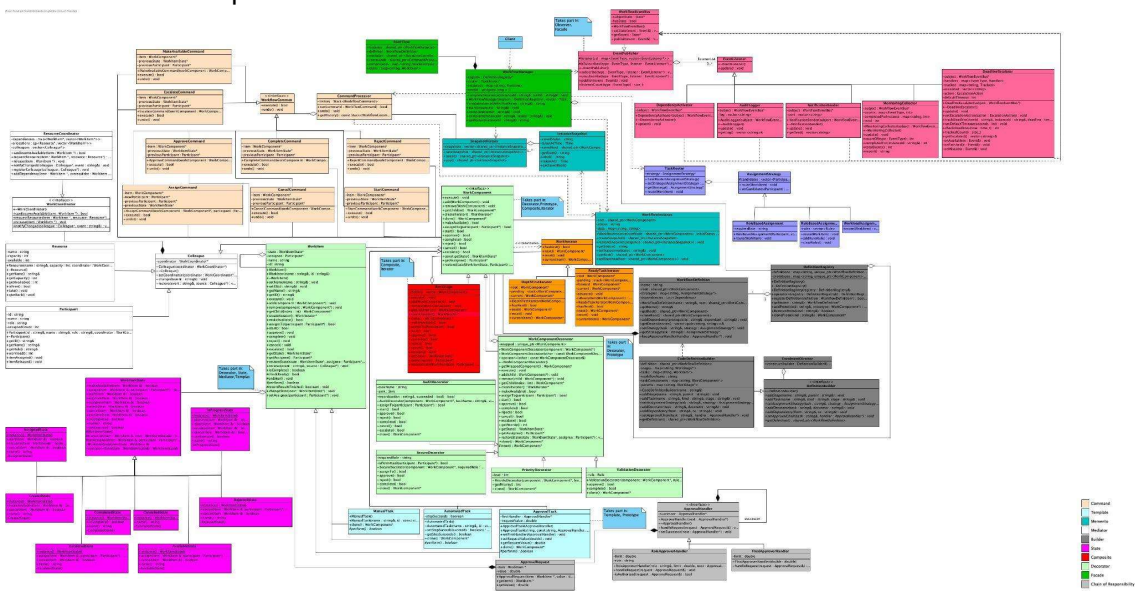
This was the initial class diagram from practical 6.
Major changes were made when compared to our final diagram updated in Task 5:
Adapter has been entirely removed as it was decided to be unnecessary and complicated to implement for a generic workflow.
Updates were made to each pattern's UML to stay consistent with our implementation.



This was our second iteration of our class diagram, which fixed all of the color issues the first one had, added some missing classes, as well as some of our relationship lines crossing each other and going over classes. This was before we completed the implementation fully, so there were many missing attributes and methods from the majority of the classes.

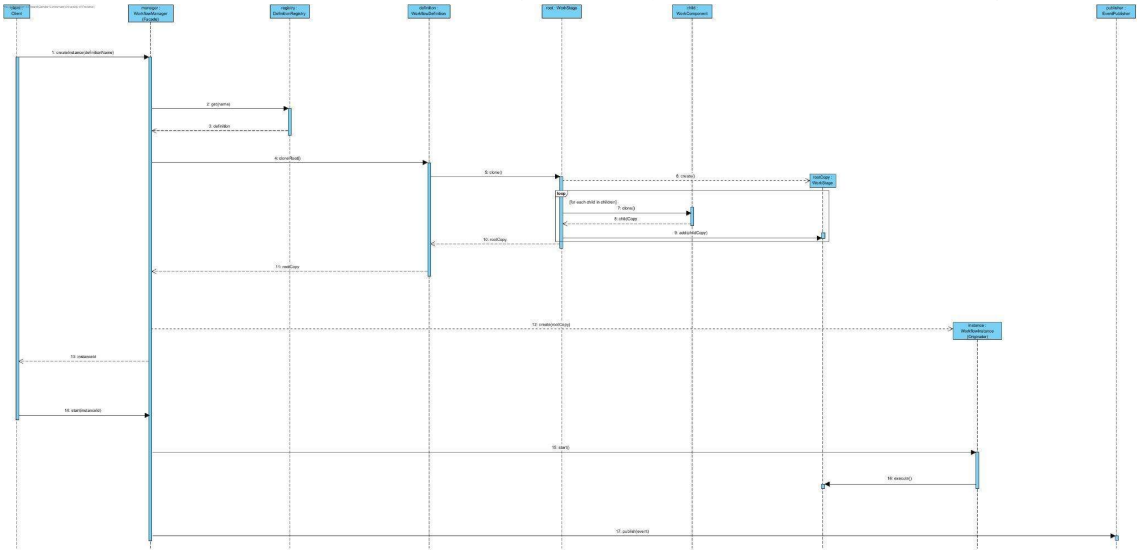


For our third and final iteration of the class diagram, we decided to wait and fix other issues before finalizing changes anymore so that everyone could be on the same page. In this iteration, we added the rest of the missing classes that were in task 4 and not in the diagram, we also fixed the rest of our relationship lines crossing each other and going through classes, we removed the adapter pattern, and finally, we updated the methods to match the .h files from the implementation.

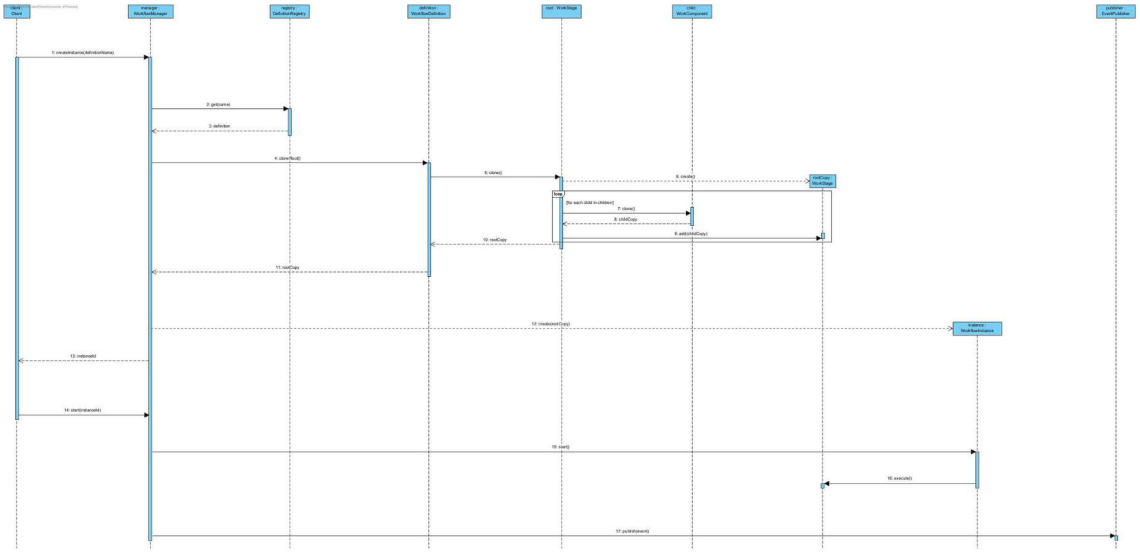


Sequence diagram changes:

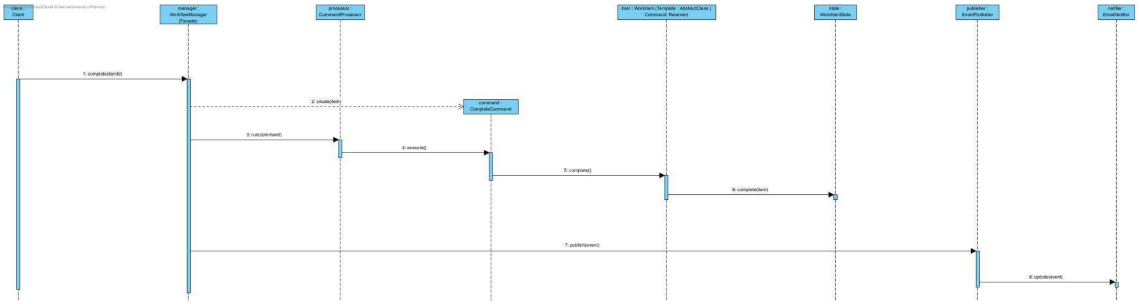
This was the first iteration of our Create and Start instance sequence diagram. This diagram had naming issues due to the names being incorrectly done in our UML class diagram.



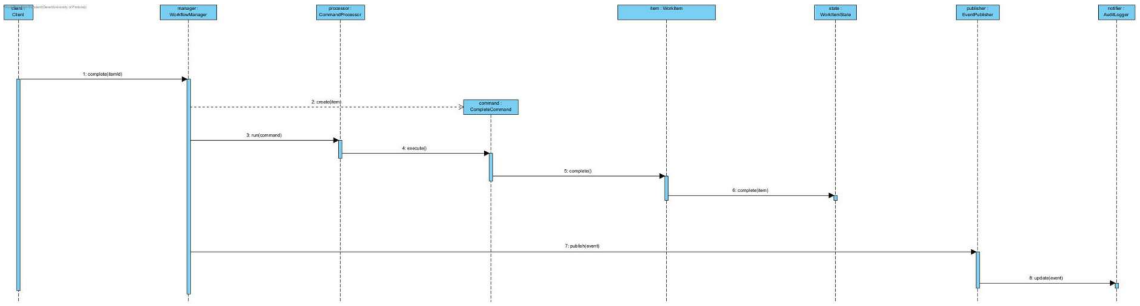
This was our second and final iteration. This diagram fixed the naming issues caused by the incorrect UML class diagram.



This was the first iteration of our Complete WorkItem sequence diagram. This diagram had naming issues due to the names being incorrectly done in our UML class diagram.

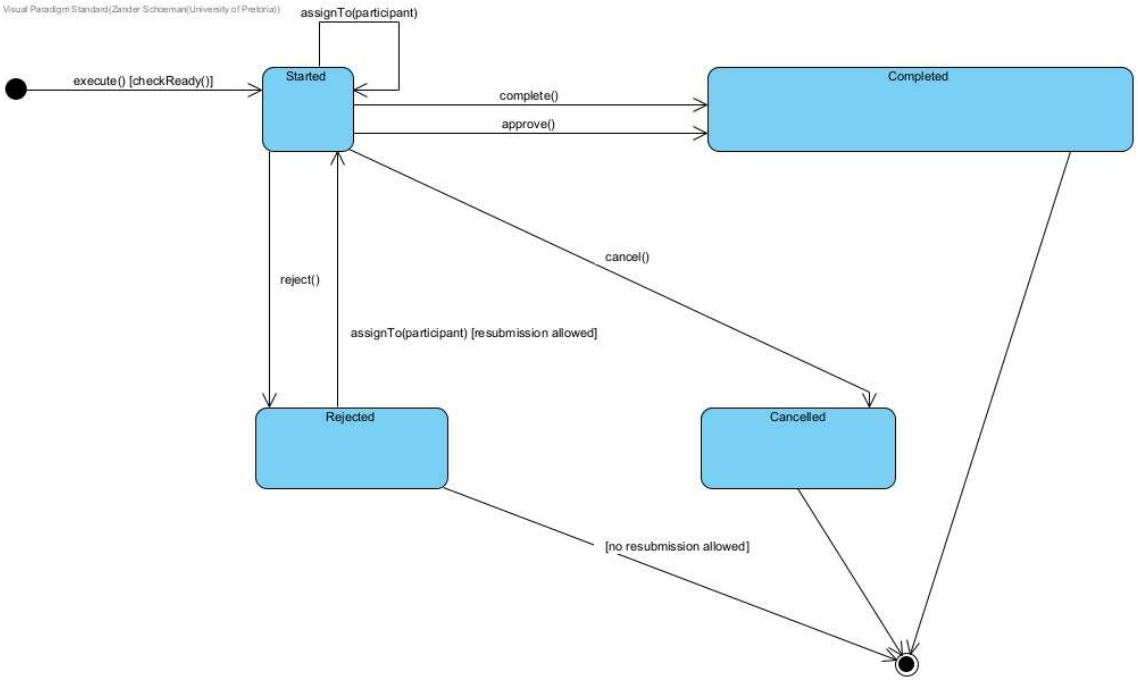


This was our second and final iteration. This diagram fixed the naming issues caused by the incorrect UML class diagram.

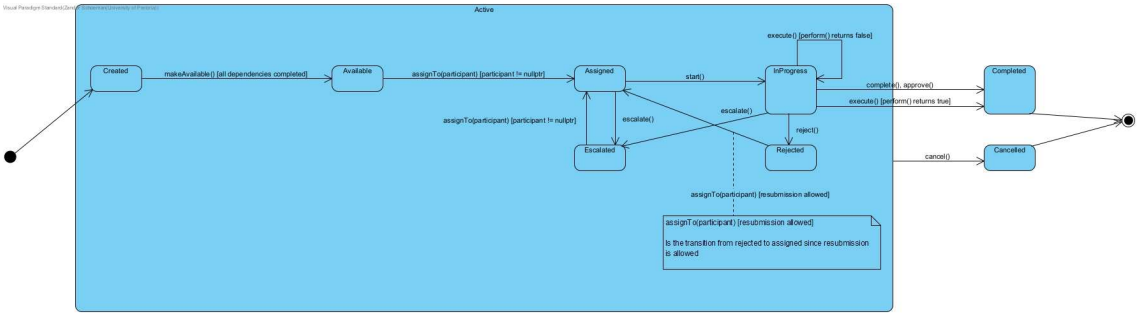


State diagram changes:

This was the first iteration of our WorkItem life cycle diagram, which was incorrect because not all the states were in the first UML class diagram and the adapter was being removed.

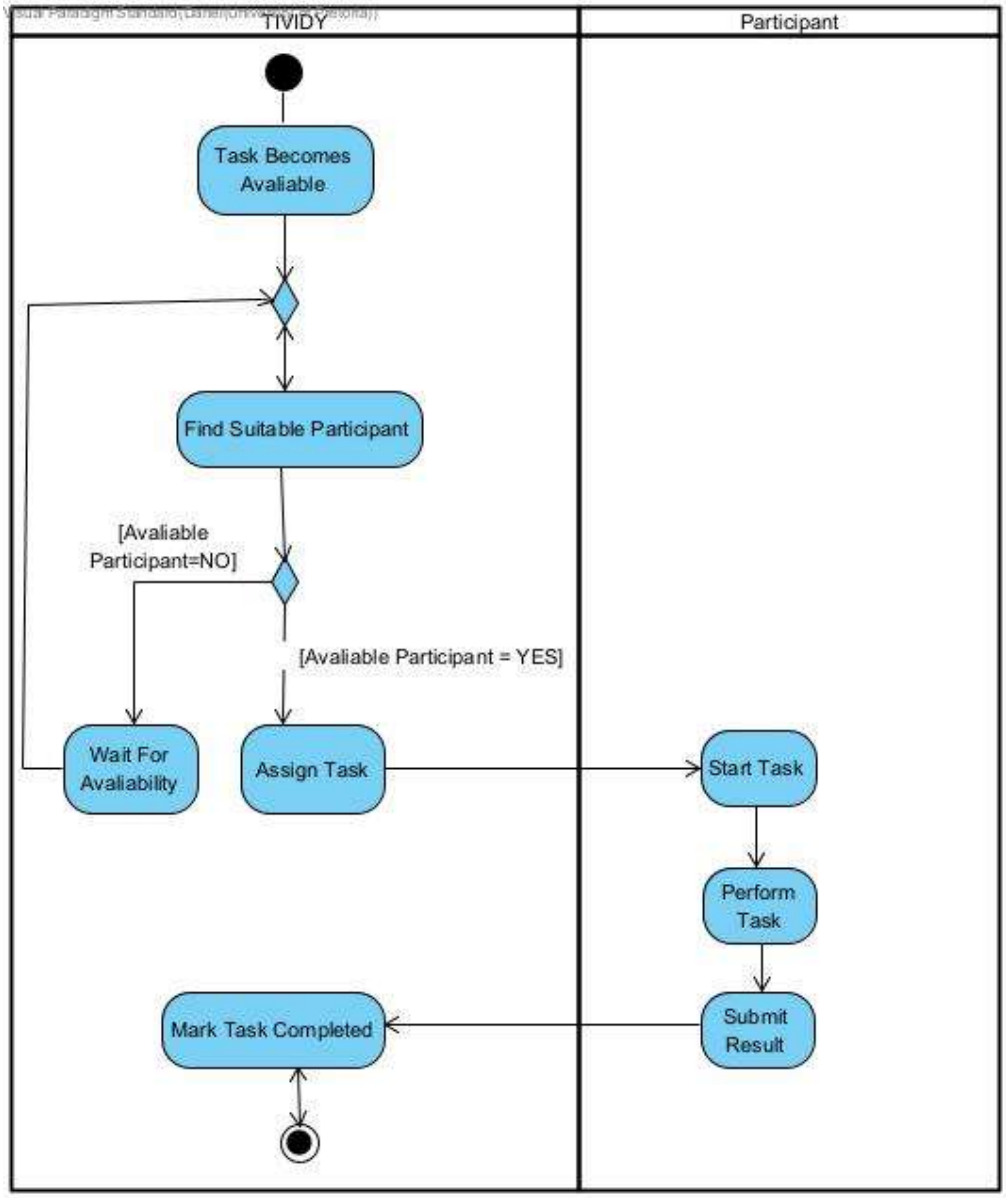


This was the second and final iteration of our WorkItem life cycle diagram. Since we added multiple additional states, the entire diagram had to be reworked to account for the new states.

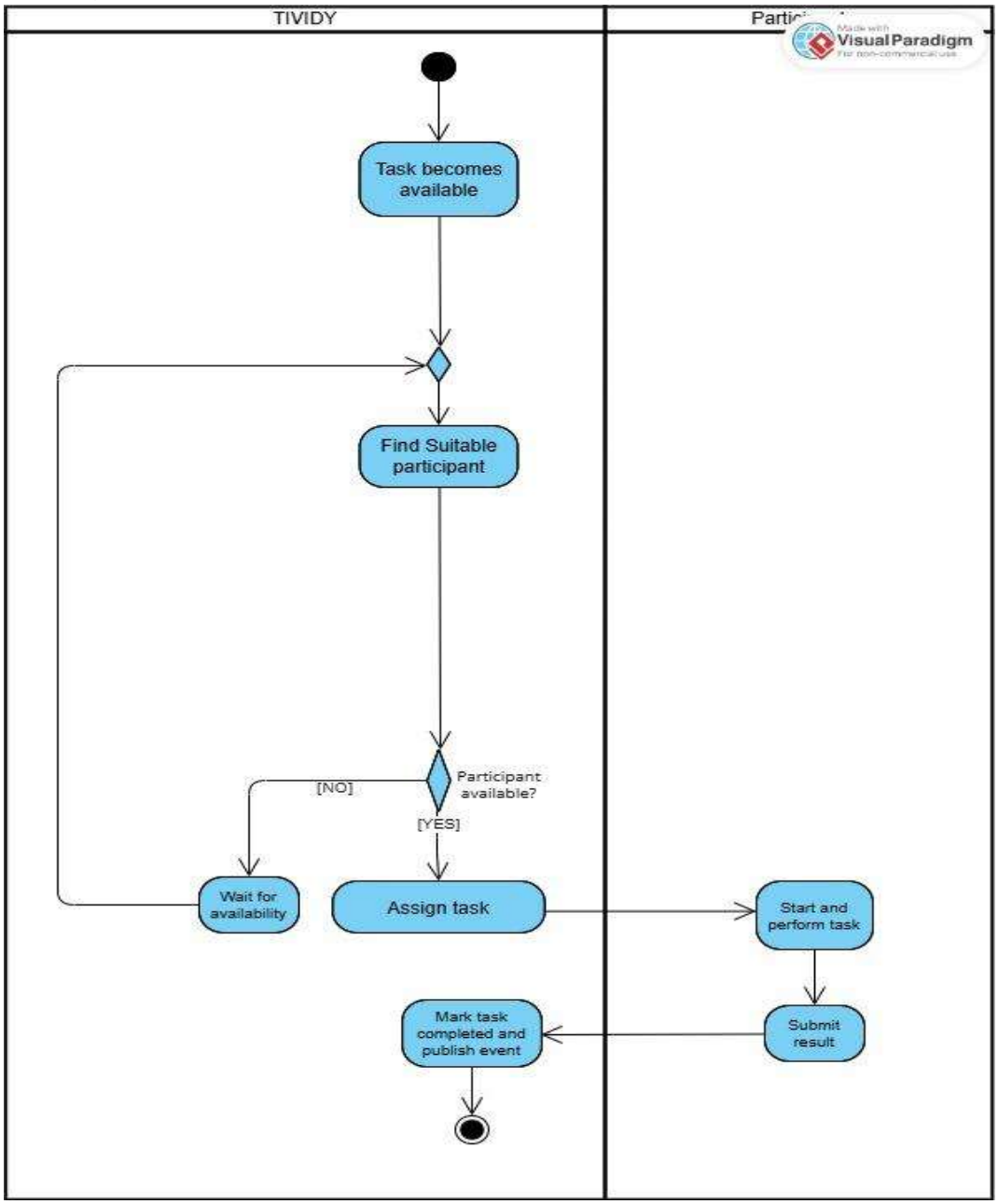


Activity diagram changes:

This was the first iteration of our Manual Task Activity diagram. This diagram had syntax errors for the decision nodes, skew lines, and activities split into multiple actions.



This was the second and final iteration of our Manual Task Activity diagram. It fixed syntax errors and merged action nodes that were incorrectly splitting activities.



Contributions:

Rafael de Gouveia - u25054903:

For this project, I handled task 4 of practical 6, which involved choosing the patterns we used based on the specifications given and how the patterns worked together, as well as documenting each pattern with the problem it solved, the participants, the way the pattern collaborated with other patterns, what might change in the patterns from generic to scenario, and why that pattern specifically was chosen over other patterns. For the project section, I helped with all the class diagram iterations as we found they needed to change, and I coded the implementation and Doxygen comments for the Mediator and the Chain of Responsibility patterns. I also handled the State, Strategy, Chain of Responsibility, Observer, Mediator, Command, and Memento patterns for the presentation slides and generated the QR code for them. Finally, I assisted in documenting the changes made to add to task 7 and updated task 3 and 6 diagrams in the documents.

Logan van der Merwe - u25025822

For this project, I tackled Task 2 of practical 6. In this task, I planned and described a specific workflow (i.e., a university enrolment workflow), then converted it into a generic workflow using the same building blocks.

In addition to laying the groundwork for the generic workflow system, I implemented the Observer and Strategy patterns, wrote the README, and provided helpful guides and documents detailing subtasks of larger documents, running instructions, et cetera. While doing this I organised the docs folder of the GitHub as well. When we decided to remove the Adapter pattern, I also helped confirm which diagrams were affected by this change and what needed to be adjusted, if needed. I contributed to the slideshow and also helped with the Docker testing.

Elme Hattingh - u25045505

For this project, I took on the role of team leader, which entailed delegating patterns to everyone in a fair manner and also helping resolve conflict, as well as doing the research on workflows, which was Task 1 of Prac 6. Furthermore, I implemented the code for the Decorator and Prototype patterns, as well as creating the Doxyfile from the Doxygen comments and fixing the errors that came up in the warnings log. I also built the Dockerfile and its supporting files in order to do the testing, and fixed the errors that appeared. I also contributed to the README.

Hanru Jansen van Rensburg - u25407725

For this project, I helped with Task 5, which is the class diagram. This includes the initial creation of the diagram and continuous updates to it throughout the whole project as changes were made to our implementation. I implemented the Memento pattern as well as the Command pattern and handled changes to be made to those patterns to accommodate minor bugs. I contributed to the slides, specifically the GitHub, Docker & CI/CD sections, to showcase how we used Git/GitHub, etc.
Also helped with Task 7, which had to do with changes to our diagrams throughout Prac 6 and the Projects.

Danel Steyn u24943802

For Practical 6, I did Task 3, creating the 3 different workflow diagrams to explain the system's processes and execution flow. I was also responsible for implementing the Builder and Facade design patterns. This also entailed some integration with the patterns involved from other teammates. I also took leadership in planning and designing the presentation, ensuring that the content was well-structured, visually consistent, and appealing.

Sergio Ferreira u25247418

For practical 6, I did Task 5 alongside Hanru and Rafael, naming all the initial patterns that were to be used for the project, creating the initial class diagram, and providing continuous updates to said diagram throughout the duration of the project. I was also responsible for the implementation of the Composite and Iterator patterns, alongside bug fixes to any other classes involving said patterns. I also handled creating the presentation slides for the Facade, Builder, Prototype, Composite, Iterator, Decorator, and Template Method patterns.

Zander Schoeman - 25097726

For practical 6, I did Task 6, which included doing two sequence diagrams and one state transition diagram, however, after we started changing functionality to accommodate the

requirements of the project, I did the new state transition diagram for the generic workflow. I also implemented the State and Template Method Design patterns, as well as designing the web GUI, which is used to demonstrate the functionality of our project.